TEXORA · LECTURE SHEET

Basic Product and Measurement

The specification sheet that defines a garment, the components of the two basic products — shirt and trouser — and the body measurements and sizing systems on which all pattern work is built.

TE-2207 · APPAREL MANUFACTURING-1 · TOPIC 2 · LEVEL 2 TERM 2

1. The specification sheet

A garment is manufactured to a document, not to an idea. The **specification sheet (spec sheet)** is the master document of a style: it states what the garment is made of, how it is constructed, and — most importantly for this topic — exactly what it must measure at every defined point, in every size, with the tolerance allowed. It travels with the order from sampling to final inspection, and any dispute about a finished garment is settled by measuring the garment against this sheet.

A complete spec sheet normally carries:

- **Style identification** — buyer, style number and name, season, size range, colour range, fabric and trims references.
- **Technical sketch** — front, back and detail views of the garment with construction call-outs (stitch types, seam allowances, special operations).
- **Points of measurement (POM)** — a list of defined measuring positions, each with a code, a description of exactly how the tape is placed, the value for every size, and the tolerance (±) permitted.
- **Construction and materials details** — bill of materials, stitch and seam specifications, labelling and packing instructions.

1.1 Example — spec sheet extract, men's casual shirt (values in cm)

POINT OF MEASUREMENT	S	M	L	XL	TOLERANCE
Chest (full, 2.5 cm below armhole)	102	107	112	117	± 1.0
Body length (from high point shoulder)	71	73	75	77	± 1.0
Shoulder width (seam to seam)	44	46	48	50	± 0.7

Sleeve length (from shoulder point)	60	61.5	63	64.5	± 0.7
Collar size (neck, buttoned)	38	40	42	44	± 0.5
Hem width (full)	100	105	110	115	± 1.0

Read the table as the industry does. The **grade** — the step between sizes — is visible in every row: chest rises by 5 cm per size, length by 2 cm, shoulder by 2 cm. The garment measurements are systematically larger than the body they fit (Topic 2, §4), the difference being *ease*. And the tolerance is part of the specification, not generosity: a chest of 106.2 cm on a size-M garment (nominal 107, tolerance ±1.0) is a failed garment.

2. Components of a shirt

The shirt is the classic teaching product of apparel manufacture because it contains nearly every basic component and operation. A standard woven shirt (Figure 1) is assembled from:

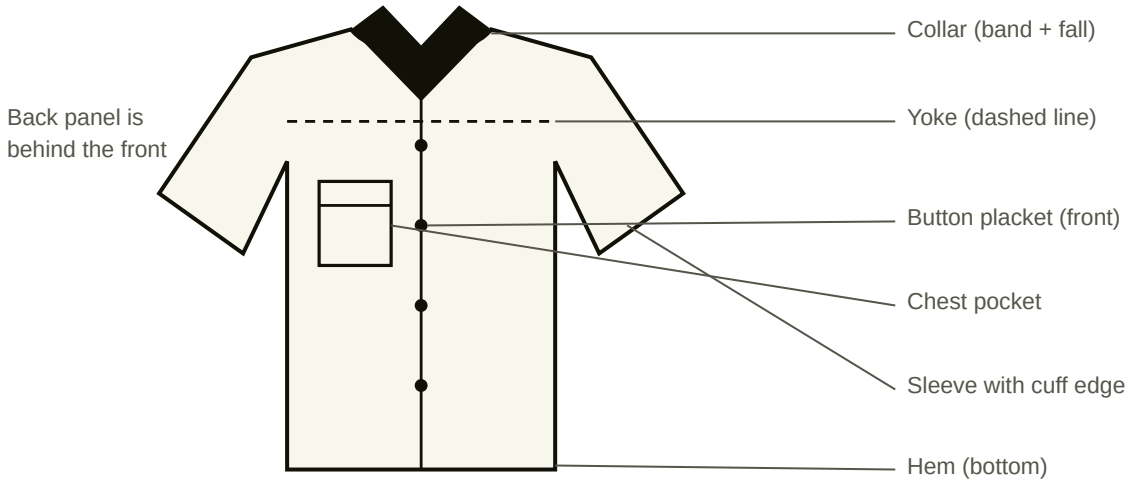


Figure 1 — Front view of a basic shirt and its principal components.

COMPONENT	FUNCTION AND COMMON TYPES
Front (×2)	The two front panels carrying the button placket and buttonholes; may carry chest pockets.
Back	One panel, often with pleats or darts for shoulder-blade ease.
Yoke	The horizontal panel across the shoulders joining front and back; single or double layer.

Collar	Two parts — the **band (stand)** that rises from the neckline and the **fall** that folds over it. Types: point, spread/cutaway, button-down, band (mandarin) collar.
Sleeve	Short or long; set-in (sewn into armhole) in shirts; raglan and drop-shoulder forms appear in casual wear.
Cuff	The band finishing the sleeve opening: single (button) cuff, double (French) cuff, convertible cuff.
Placket	The reinforced front opening carrying buttons and buttonholes; sleeve placket opens the cuff seam line.
Pocket	Chest patch pocket, with or without flap; shape and position fixed by the spec.

3. Components of a trouser

The trouser is the second basic product. Its parts are cut as pairs — left and right legs — and symmetry between them is a permanent quality concern.

COMPONENT	FUNCTION AND COMMON TYPES
Front (×2) and back (×2)	The four leg panels; backs are cut larger across the seat. Fronts may carry pleats (one or two) or be flat-front.
Waistband	The firm band closing the top; plain, extended (with a tab), or with belt loops and elastic inserts.
Fly	The front opening: zip fly (most common), button fly; built with fly piece and fly shield.
Pockets	Front: slanted (cross), scoop or straight pockets; back: welt, flap or patch pockets.
Crease line	The pressed fold down the centre of each leg, aligned to the grain in cutting.
Hem / bottom	Plain turned-up hem, or a cuffed (turn-up) bottom; length set by the inside-leg measurement.
Loops, waistband lining, pocket bags	Secondary components that finish and support the main ones; cut from shell fabric or lining/pocketing material.

Comparing the two products teaches a general rule: every garment, however complex, is a set of **main components** (seen outside), **secondary components** (openings, closures, supports) and **hidden components** (interlinings, pocket bags, linings). The pattern maker of Topic 3 must produce a correct piece for every one of them.

4. Body measurement

Patterns are not drawn from imagination; they are drawn from the dimensions of the human body. Body measurement follows standard rules: the subject stands erect in light clothing, the tape is held snug but not tight, and each measurement is taken at a defined anatomical point so that two measurers obtain the same figure. The principal measurements are shown in Figure 2.

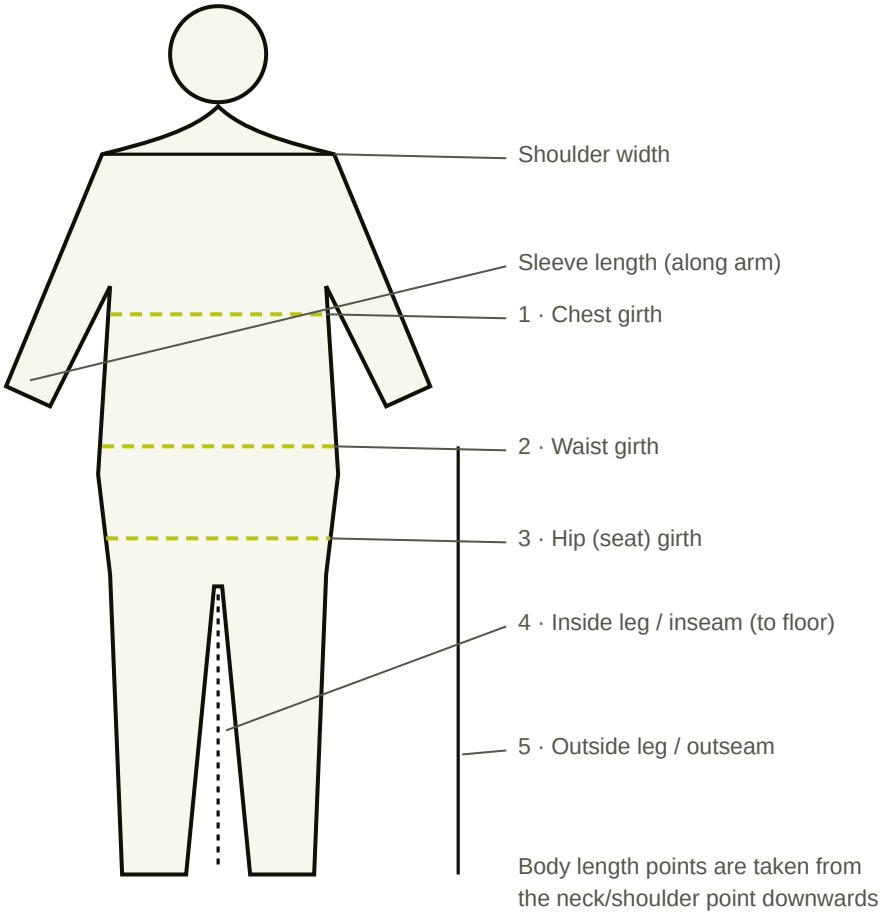


Figure 2 — The principal body measurements: girths (1–3, shown by the green dashed lines around the body) and lengths (4–5 and shoulder/sleeve).

4.1 Standard body measurements (example charts, cm)

GENTS — SIZE	S	M	L	XL
Chest girth	92	97	102	107
Waist girth	76	81	86	92
Hip (seat) girth	94	99	104	109
Arm length (shoulder to wrist)	60	61.5	63	64.5

Inside leg	76	78	80	81

LADIES — SIZE	10	12	14	16
Bust girth	84	88	92	97
Waist girth	66	70	74	79
Hip girth	90	94	98	103
Back length (neck to waist)	40	41	42	43
Arm length	58	59	60	61

Compare the gents' body chart with the shirt spec of §1.1: body chest 97 cm (size M) becomes garment chest 107 cm — the 10 cm difference is **ease**: wearing ease that lets the body move and breathe, plus design ease that gives the garment its intended silhouette. Ease is never guessed; it is part of the style's design.

5. Sizing systems and sizing ranges

A **sizing system** is the rule that names sizes and ties them to body dimensions. No single world system exists, so a technologist must convert between them daily:

- **Letter sizes** — XS, S, M, L, XL, XXL…; simple but imprecise, common for knitwear and casual wear.
- **Body-measurement sizes** — the size is the body measurement itself: a man's shirt by neck size (38, 40, 42 cm, or 15, 15½, 16 in), trousers by waist (30, 32, 34 in).
- **Numbered ladies' sizes** — UK 8–22 and US 2–20 follow related sequences offset by two (UK 12 ≈ US 8); European sizes run higher again (UK 12 ≈ EU 40).
- **Height systems** — children's clothing is sized by body height (92, 104, 116 cm…), and men's tailoring adds height letters (S/R/L for short, regular, long) to the chest size.

MEN'S SHIRT COLLAR	UK / US (IN)	14½	15	15½	16	16½	17
	EU (cm)	37	38	39–40	41	42	43

The **sizing range** of an order is the set of sizes to be produced — for example S–XXL in a stated ratio (such as 1:2:2:1 across S:M:L:XL for a four-size order). The range decides how many patterns must be graded (Topic 3) and how the marker is planned (Topic 4), and the ratio decides how many garments of each size enter each cutting.

6. Glossary

TERM	MEANING
Point of measurement (POM)	A defined position and method for measuring a garment, listed on the spec sheet with values and tolerance.
Ease	The amount by which a garment measurement exceeds the body measurement: wearing ease for movement, design ease for silhouette.
Tolerance	The permitted deviation (±) from a spec value within which a garment still passes.
Grade (size step)	The fixed difference between consecutive sizes at a given measurement point.
Size ratio	The proportion of garments ordered in each size of the range, used in marker and cutting planning.

7. Key points and summary

- The spec sheet — sketch, materials and POM table with tolerances — is the contract every garment is measured against.
- Shirt components: fronts, back, yoke, collar (band + fall), sleeve, cuff, placket, pocket; trouser components: leg panels, waistband, fly, pockets, crease, hem.
- Body measurements are girths (chest/bust, waist, hip) and lengths (shoulder, sleeve, inside leg), taken at defined anatomical points under standard rules.
- Garment measurement = body measurement + ease; sizing systems (letters, body-measure, numbered, height) must be converted between markets.
- The sizing range and ratio of an order drive grading and marker planning in the topics that follow.

8. Practice

Answer in your own words; brief model answers follow.

1. From the spec extract in §1.1, a size-L shirt measures chest 110.4 cm. Pass or fail? Show the limits you used.
2. Using the charts in §4.1, calculate the chest ease of the size-M shirt and state the two kinds of ease it contains.
3. A buyer orders sizes S:M:L:XL in the ratio 1:2:2:1 for 1,200 pieces. How many pieces of each size?
4. Convert a UK/US collar size of 16½ in to the EU size, using the table in §5, and convert a garment length of 30 in to centimetres (1 in = 2.54 cm).

5. Name the two parts of a shirt collar and the component that joins the front and back panels across the shoulders.
6. A trouser pattern set is found to contain two left fronts and no right front. Which general component rule has been broken, and at which stage should it have been caught?

Answer key

1. Nominal L chest = 112 cm, tolerance ±1.0 → acceptable range 111.0–113.0 cm. 110.4 cm is below the range: fail.
2. Garment chest 107 − body chest 97 = 10 cm ease, made of wearing ease (movement, breathing) plus design ease (the shirt's intended fullness).
3. Ratio parts = 1+2+2+1 = 6; one part = 200 pieces. S 200, M 400, L 400, XL 200.
4. 16½ in ≈ EU 42. 30 in × 2.54 = 76.2 cm.
5. The band (stand) and the fall; the yoke joins front and back across the shoulders.
6. The rule that garments are built from mirrored pairs/symmetric component sets. It should be caught at pattern checking — and certainly before cutting, when the marker is planned.
